

W07 · M07 · 20 MINIT / MINUTES · Hari 2, 14:35 hingga 14:55

Hands-On SQA-AI Assistant Workshop

AI audit, CAPA and Test Summary Report

Audit, CAPA dan Laporan Ringkasan Ujian dibantu AI

Audit your outputs folder against the ISO/CMMI-lite checklist, raise CAPA for non-conformities and draft the D14/D16 Test Summary with a release recommendation.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

Before you start / Sebelum mula

YOU WILL / ANDA AKAN

1. Run /audit on the RTM and test cases (Plan agent)
2. Switch to Build and save the audit to outputs/
3. Run /capa on the saved audit, then /test-summary
4. Decide GO, NO-GO or CONDITIONAL as a team

1. Jalankan /audit pada RTM dan kes ujian (ejen Plan)
2. Tukar ke Build dan simpan audit ke outputs/
3. Jalankan /capa pada audit yang disimpan, kemudian /test-summary
4. Putuskan GO, NO-GO atau CONDITIONAL sebagai pasukan

CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)
- Outputs from earlier workshops exist in outputs\ (or copy them from example-output)

Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

OUTPUT

outputs/m07-audit.md, outputs/m07-capa.md, outputs/m07-test-summary.md

W07 · STEP 1 OF 7 · LANGKAH 1

Run /audit in Plan mode on the RTM and test cases. It prints conformities, non-conformities (NC Major and Minor) and evidence.

Jalankan /audit dalam mod Plan pada RTM dan kes ujian. Ia memaparkan pematuhan, ketidakpatuhan (NC Major dan Minor) dan bukti.

TYPE IN OPENCODE (PLAN)

Salin / Copy

/audit outputs/m02-rtm.csv dan outputs/m04-testcases.csv

OC | Audit KRISAv2 dengan Template

RTM-guide.md seksyen 5.

Jadual Kiraan

Patuh	NC Major	NC Minor	OFI	TB
1	2	3	2	15

Catatan: kiraan NC Major, NC Minor dan OFI mengikut bilangan penemuan AUD; TB mengikut bilangan perkara AU yang berada di luar skop audit.

Rujukan

- PPrISA05 Pelan Pengurusan Kualiti: [PPrISA05 PDF \(../references/pprisa-2.0/templates/pdf/PPrISA05-Pelan Pengurusan Kualiti.pdf\)](#)

- PPrISA 2.0, 5.4.2 ii b) Penilaian Kualiti Serahan Projek: [halaman 94 \(../references/pprisa-2.0/PPrISA 2.0_Versi Beta Februari 2025.pdf#page=94\)](#)

- KRISAv2 Bab 1, 1.6.4 Jaminan Kualiti Perisian (SQA): [BAB1-PERANCANGAN.pdf, halaman 19 \(../references/krisa-v2-beta-2026/BAB1-PERANCANGAN.pdf#page=19\)](#)

- Sokongan keterjejukan: KRISAv2 Bab 2, 2.11 RTM Templat T2.8 [F1.8: BAB2-FASA-PERMULAAN.pdf, halaman 56 \(../references/krisa-v2-beta-2026/BAB2-FASA-PERMULAAN.pdf#page=56\)](#)

- Senario dan kes ujian: KRISAv2 Bab 6, 6.7 [F5.2: BAB6-FASA-PENGUJIAN-PENERIMAAN.pdf, halaman 9 \(../references/krisa-v2-beta-2026/BAB6-FASA-PENGUJIAN-PENERIMAAN.pdf#page=9\)](#)

Untuk simpan, salin ke outputs/m07-audit.md dan tukar pautan kepada ../../references/....

Semak manusia / Human check

1. Buka setiap fail bukti yang disebut dan sahkan ia wujud dan benar-benar menyokong klasifikasi.

2. Klasifikasi NC Major dan NC Minor diputuskan oleh juruaudit manusia berdasarkan kriteria objektif, bukan pendapat AI.

3. Setiap NC dimaklumkan kepada auditee dan mempunyai CAPA yang dirancang melalui /capa.

Plan · Big Pickle · 5m 14s

Plan · Big Pickle OpenCode Zen

C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab

89.1K (45%) ctrl+p commands

Audit KRISAv2 dengan Template

Context

89,148 tokens

45% used

\$0.00 spent

LSP

LSPs are disabled

Getting started

OpenCode includes free models so you can start immediately.

Connect from 75+ providers to use other models, including Claude, GPT, Gemini etc

Connect provider /connect

/~/trainee-demo\sqa-krisa-bengkel\lab: main

• OpenCode 1.18.30

W07 · STEP 2 OF 7 · LANGKAH 2

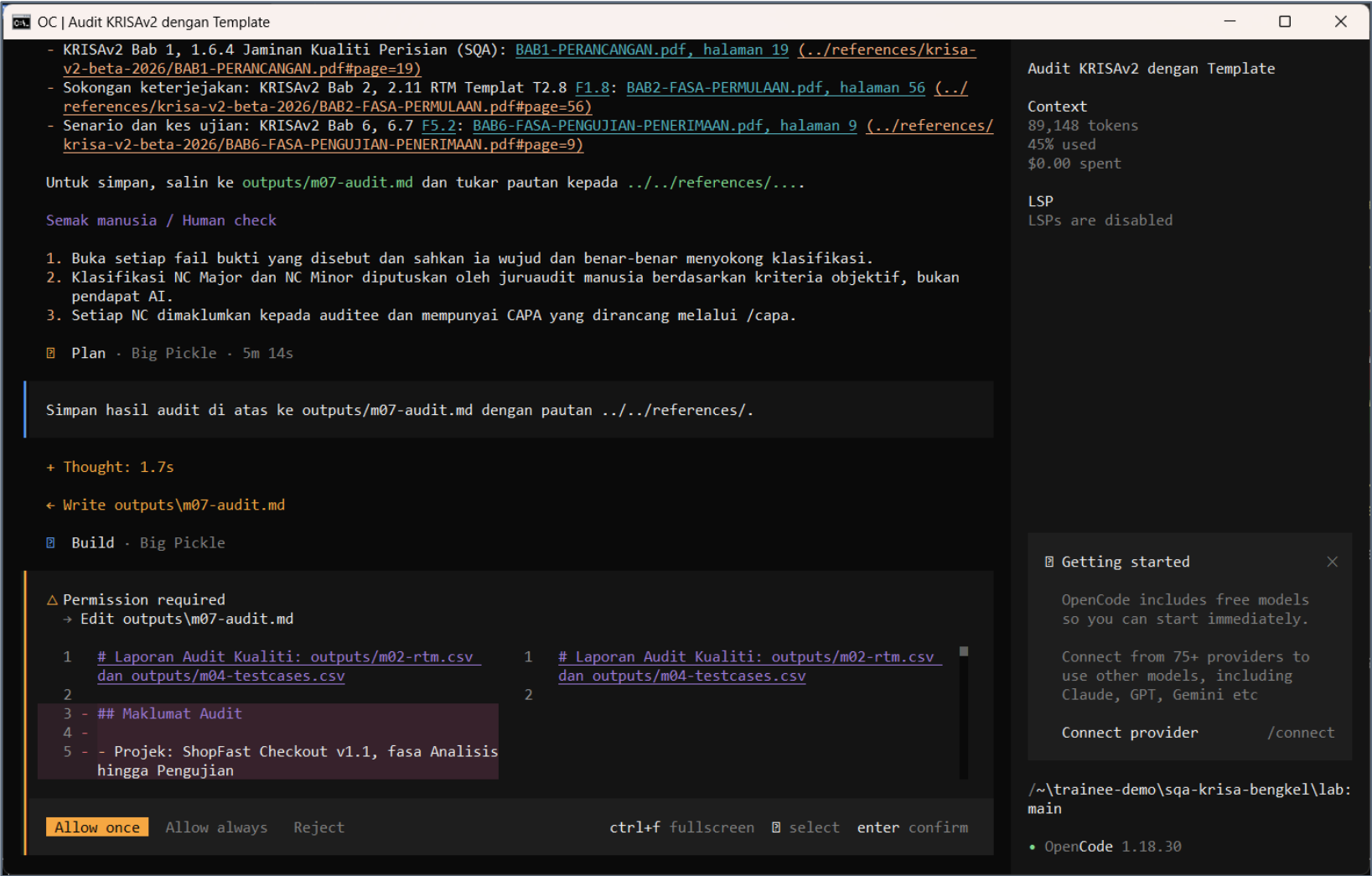
Press Tab to switch to Build and ask the assistant to save the audit. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Tekan Tab untuk tukar ke Build dan minta pembantu menyimpan audit. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

Salin / Copy

Simpan hasil audit di atas ke outputs/m07-audit.md dengan pautan ../../references/.



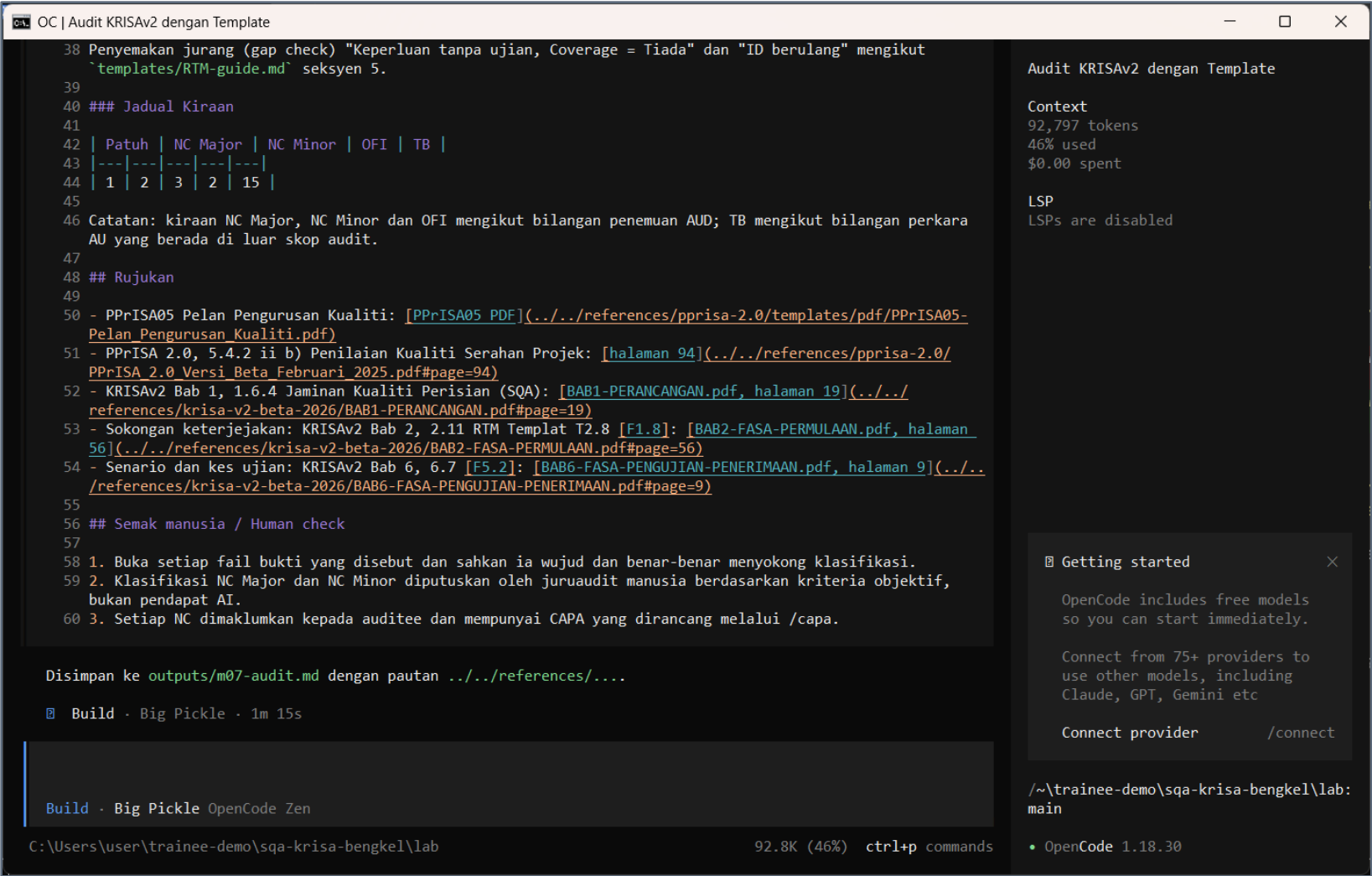
Allow once. The audit is saved as evidence in outputs/m07-audit.md.

Allow once. Audit disimpan sebagai bukti dalam outputs/m07-audit.md.

TYPE IN OPENCODE

Salin / Copy

Allow once (Enter)



W07 · STEP 4 OF 7 · LANGKAH 4

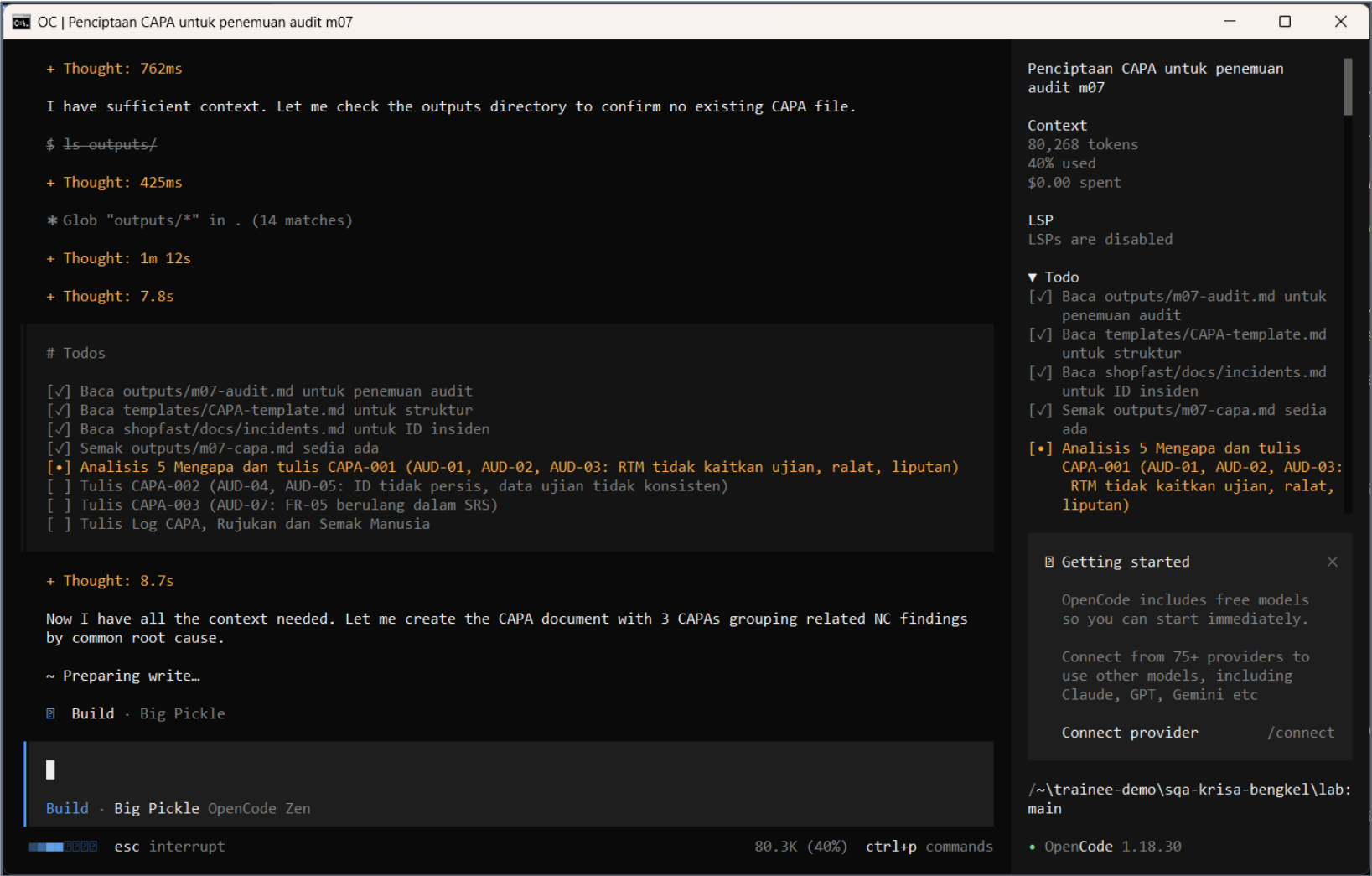
Run /capa on the saved audit file. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /capa pada fail audit yang disimpan. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

Salin / Copy

/capa semua NC dalam outputs/m07-audit.md



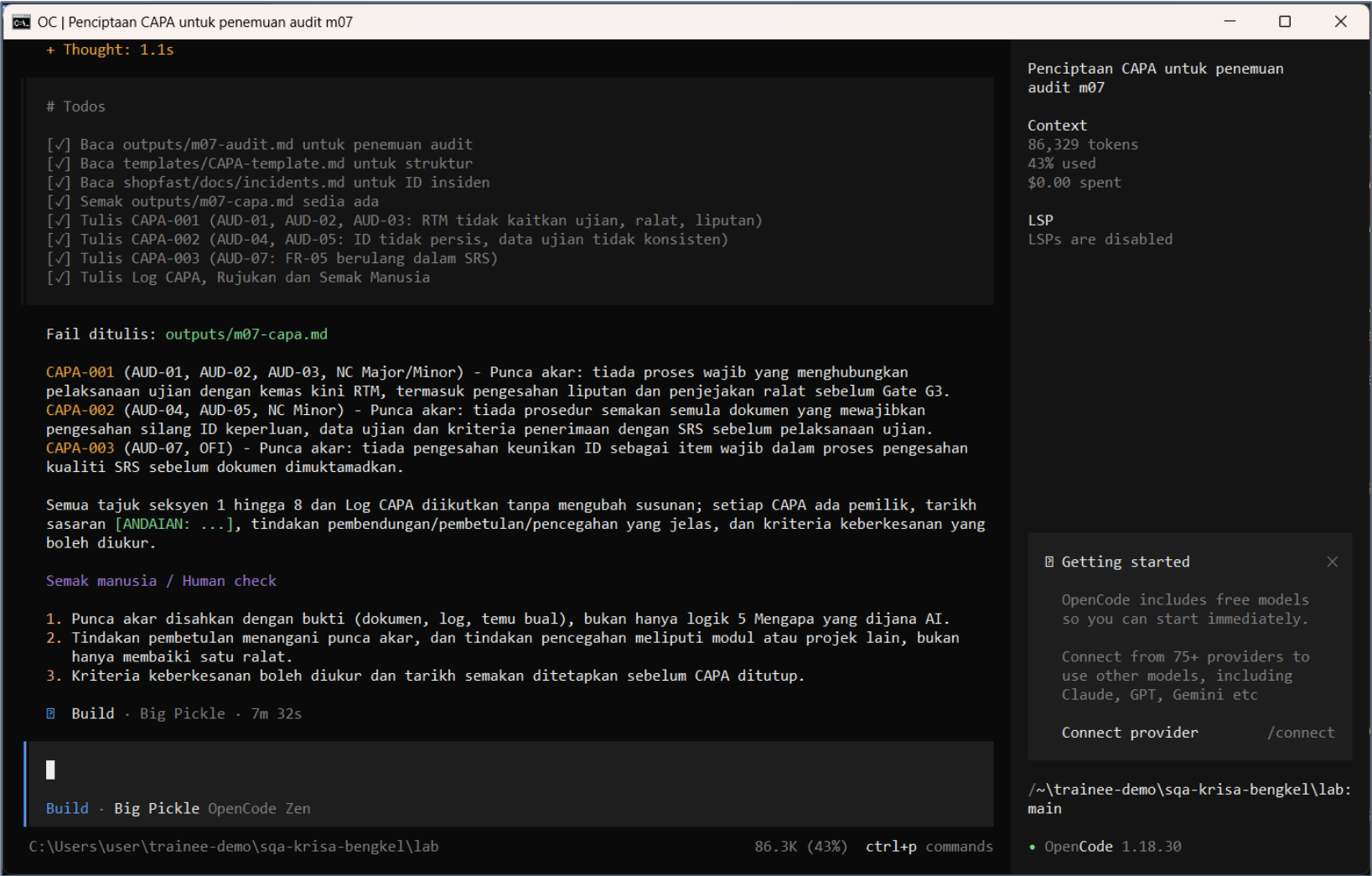
Allow once. Each NC gets root cause (5 Whys), correction, prevention, owner and date.

Allow once. Setiap NC mendapat punca akar (5 Mengapa), pembetulan, pencegahan, pemilik dan tarikh.

TYPE IN OPENCODE

Salin / Copy

Allow once (Enter)



W07 · STEP 6 OF 7 · LANGKAH 6

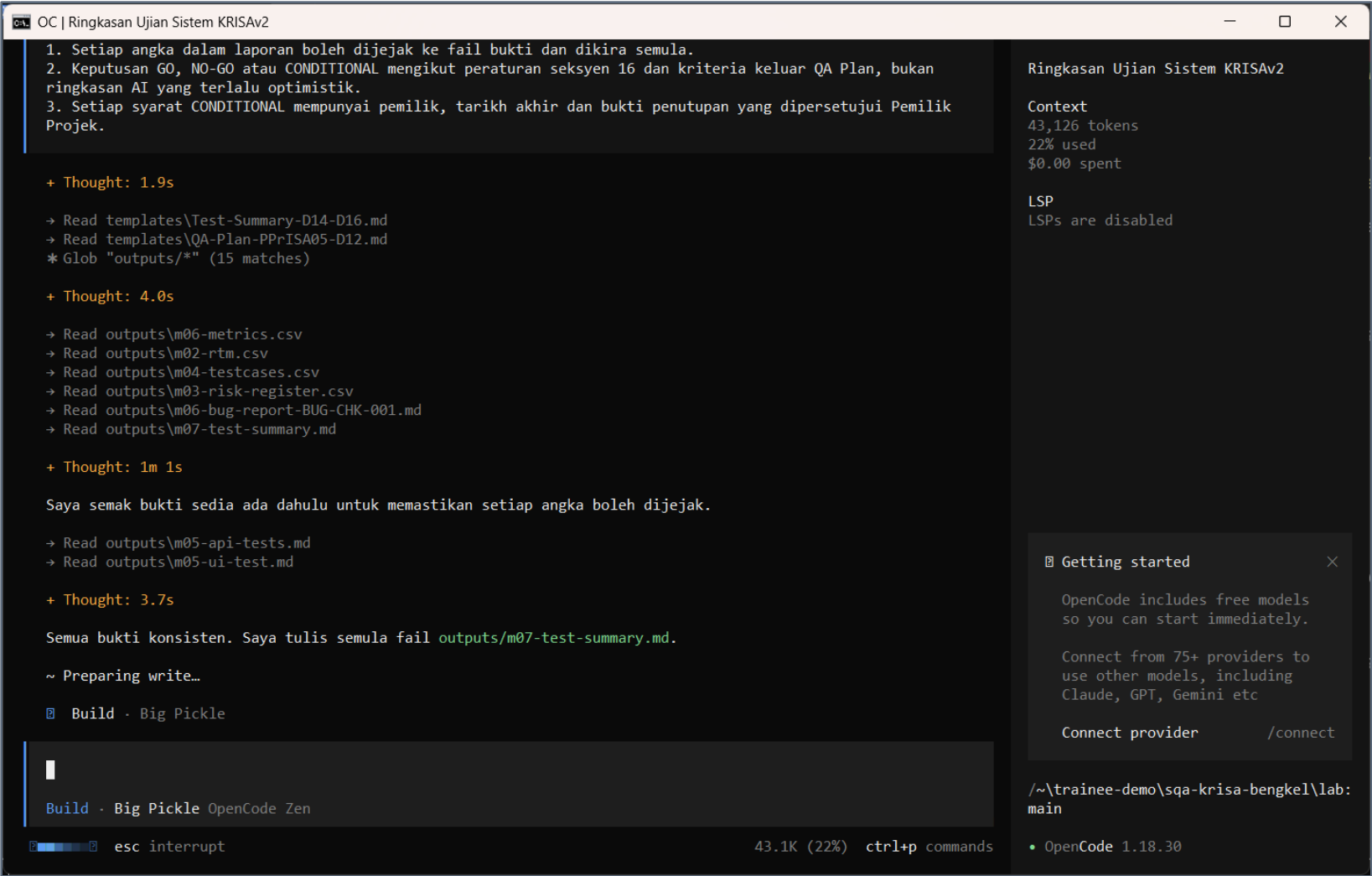
Run /test-summary with the test period and build. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /test-summary dengan tempoh ujian dan binaan. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

Salin / Copy

/test-summary Ujian sistem 14 hingga 15 September 2026, binaan 1.0.0, penguji 4 orang



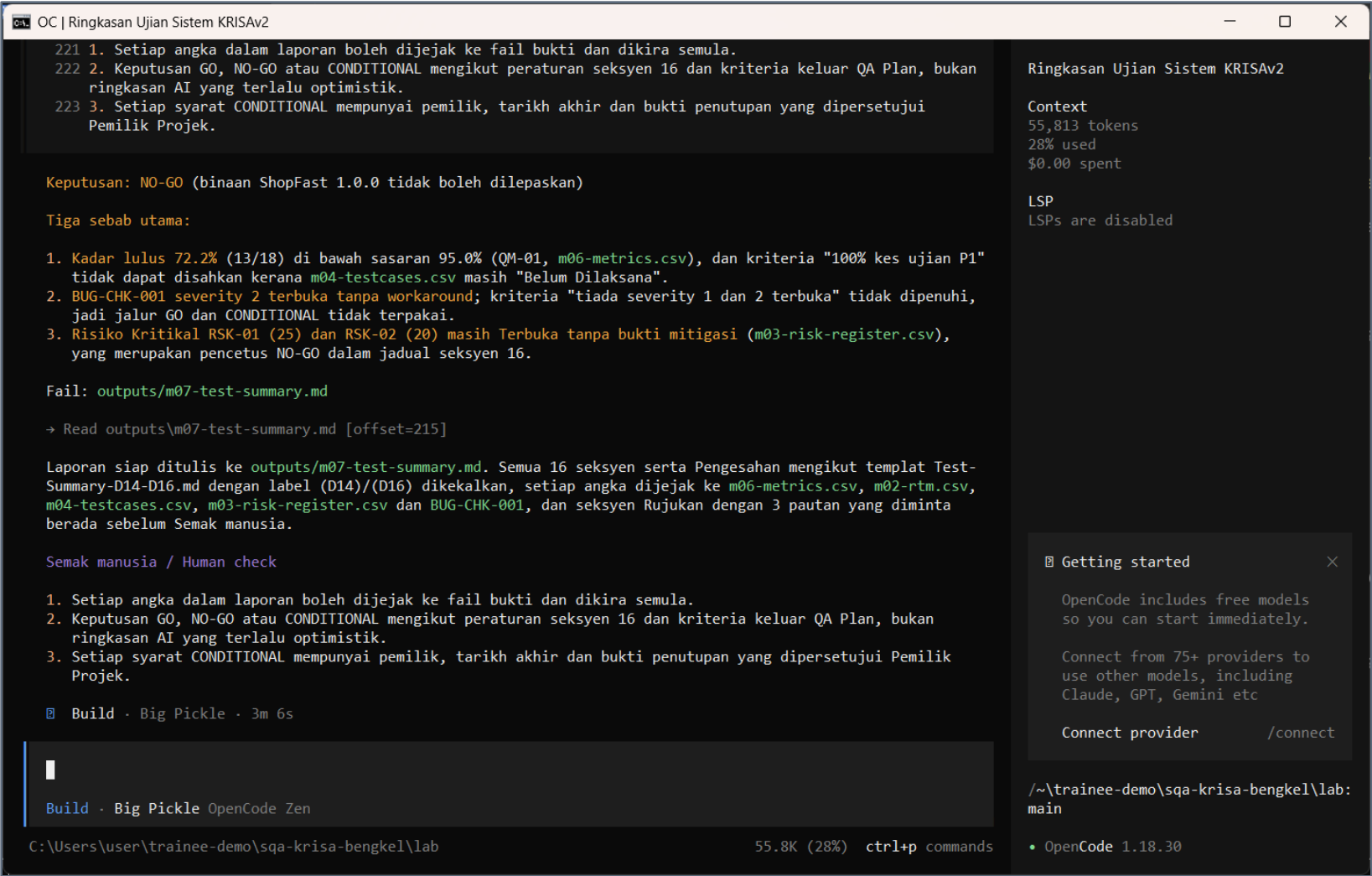
Allow once. Read the recommendation, then decide as a team.

Allow once. Baca cadangan, kemudian putuskan sebagai pasukan.

TYPE IN OPENCODE

Salin / Copy

Allow once (Enter)



Human check before you accept the output / Semakan manusia

Each non-conformity cites evidence (file and line)

Setiap ketidakpatuhan memetik bukti (fail dan baris)

Exit criteria use KRISAv2 Bab 6 6.8 numbers (95% pass, severity 1 and 2 closed)

Kriteria keluar menggunakan angka KRISAv2 Bab 6 6.8 (95% lulus, severity 1 dan 2 ditutup)

The release decision is made and signed by a person

Keputusan keluaran dibuat dan ditandatangani oleh manusia

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

If something goes wrong / Jika ada masalah

opencode is not recognised

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

Scripts are disabled on this system

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

The model is slow or stops

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

Permission required dialog

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

File or command not found

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

Answer uses Indonesian words or dashes

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

Network or proxy error

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting opencode

Port 3000 already in use

`$env:PORT = 3001` before npm start, and use `http://localhost:3001`

Abbreviations / Singkatan (1 of 3)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data. <i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i>
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen. <i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>
CAPA	Corrective and Preventive Action	Fixing a problem that has happened, then changing the process so it does not happen again. <i>Tindakan Pembetulan dan Pencegahan. Membaiki masalah yang berlaku dan mengubah proses supaya ia tidak berulang.</i>
CMMI	Capability Maturity Model Integration	The newer model that replaced CMM. Used to improve and appraise how an organisation works. <i>Model baharu yang menggantikan CMM. Digunakan untuk menambah baik dan menilai cara organisasi bekerja.</i>
FAT	Factory Acceptance Test	The formal test with the customer before the system is installed at the live site. <i>Ujian Penerimaan Kilang. Ujian formal bersama pelanggan sebelum sistem dipasang di tapak sebenar.</i>
IEC	International Electrotechnical Commission	An international standards body that publishes many software standards together with ISO. <i>Badan piawaian antarabangsa yang menerbitkan banyak piawaian perisian bersama ISO.</i>

Abbreviations / Singkatan (2 of 3)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
IEEE	Institute of Electrical and Electronics Engineers	A professional body that publishes software engineering standards such as IEEE 730 and IEEE 829. <i>Badan profesional yang menerbitkan piawaian kejuruteraan perisian seperti IEEE 730 dan IEEE 829.</i>
ISO	International Organization for Standardization	Publishes international standards such as ISO 9001 and ISO/IEC 25010. <i>Pertubuhan Antarabangsa bagi Pemiawaian. Menerbitkan piawaian seperti ISO 9001 dan ISO/IEC 25010.</i>
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition. <i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>
MyIPO	Perbadanan Harta Intelek Malaysia	Intellectual Property Corporation of Malaysia, the host of this training venue. <i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>
NC	Nonconformity	An audit finding where the work does not meet a requirement. Rated Major or Minor. <i>Ketidakakuran. Penemuan audit apabila kerja tidak memenuhi keperluan. Dikelaskan sebagai Major atau Minor.</i>
PAT	Provisional Acceptance Test	The KRISA acceptance test done before provisional acceptance of the system, including performance checks. <i>Ujian Penerimaan Sementara dalam KRISA sebelum sistem diterima secara sementara, termasuk semakan prestasi.</i>

Abbreviations / Singkatan (3 of 3)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates. <i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>
RTM	Requirements Traceability Matrix	A table linking each requirement to its design, tests and defects, so nothing is missed. <i>Matriks Kebolehkesanan Keperluan. Jadual yang memautkan setiap keperluan kepada reka bentuk, ujian dan kecacatan supaya tiada yang tertinggal.</i>
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements. <i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>
TUI	Text User Interface	A full-screen program that runs inside a terminal window, like opencode. <i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>
UAT	User Acceptance Test	Users try the system to confirm it meets their needs before they accept it. <i>Ujian Penerimaan Pengguna. Pengguna mencuba sistem untuk memastikan ia memenuhi keperluan sebelum menerimanya.</i>

Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)

[Bab 6, 6.10 Laporan Ujian Penerimaan \(UAT dan PAT\) \(p.26\)](#)

[Bab 7, 7.6 Ujian Penerimaan Akhir \(FAT\) \(p.7\)](#)

[Bab 7, Laporan Penamatan Ujian \(Jadual 7.5, ISO/IEC/IEEE 29119\) \(p.9\)](#)